
BioKG-Align: Technical Whitepaper

Problem Definition, Dataset Construction, and Evaluation Methodology

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Abstract

This whitepaper provides the full technical specification for the BioKG-Align NeurIPS 2026 competition, complementing the competition proposal with formal problem definitions, complete dataset construction methodology, and detailed evaluation protocols that could not be accommodated within the proposal page limits. BioKG-Align casts ontology matching as typed link prediction and OWL expression generation over ontology-derived biomedical knowledge graphs, exposing the problem through interfaces familiar to the knowledge graph learning and neuro-symbolic AI communities. We describe the OWL-to-graph projection pipeline, the OWL 2 RL Datalog extraction, the reference alignment construction protocol, the pool-based candidate ranking task design, and the full family of evaluation metrics, including Macro Preferred Relation-Aware MRR, Hierarchy-Aware Typed nDCG@10, and graph-edit-distance F-measure for the complex track, with formal definitions, worked examples, and baseline results on dataset release v0.2.0-rc3 (30,216 test queries across three ontology-pair tasks and 15,636 reference logical definitions across three phenotype ontologies).

1 Introduction

Biomedical ontologies are the structured vocabularies underpinning electronic health records, genomic databases, clinical decision support systems, and large-scale biomedical AI. Resources such as SNOMED CT [Donnelly et al., 2006], the Foundational Model of Anatomy (FMA) [Rosse and Mejino, 2003], the NCI Thesaurus (NCIt) [Golbeck et al., 2003], the Disease Ontology (DOID) [Schriml et al., 2012], and the Orphanet Rare Disease Ontology (ORDO) [Vasant et al., 2014] collectively span hundreds of thousands of biomedical concepts; yet they were developed independently, with distinct scopes, granularity assumptions, and internal semantics. Ontology matching [Euzenat and Shvaiko, 2013] addresses this fragmentation by identifying typed semantic correspondences between entities across ontologies.

BioKG-Align is a competition that lowers the entry barrier to this problem for the machine learning and neuro-symbolic AI communities by projecting OWL-based biomedical ontologies into graph-learning-ready knowledge graphs and framing alignment as typed link prediction over fixed candidate sets. The competition provides three progressively richer information layers (graph triples, node

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properties, and Datalog rules derived from OWL 2 RL axioms), enabling methods from standard KG embedding to neuro-symbolic reasoning.

This whitepaper provides the technical specification deferred from the competition proposal, covering: (i) formal problem definitions for the main and complex tracks (§2); (ii) the complete dataset construction pipeline, including source acquisition, normalization, graph and Datalog projection, reference alignment construction, pool-based query design, candidate generation, and leakage controls (§3); (iii) complete dataset statistics for release v0.2.0-rc3 (§4); (iv) the full evaluation methodology with formal metric definitions (§5); and (v) baseline implementations and results (§6).

2 Problem Definition

2.1 Shared Setup and Notation

Let $\mathcal{O} = \{O_1, \dots, O_k\}$ be a collection of OWL 2 ontologies. Each ontology O_i induces a projected graph $G_i = (V_i, E_i)$ under the projection procedure described in §3.3. The *unified competition graph* is

$$G = \left(\bigcup_i V_i, \bigcup_i E_i \cup E_{\text{anchor}} \right),$$

where E_{anchor} contains public cross-ontology training anchors (§3.5). Node properties are collected into a matrix $X : \bigcup_i V_i \rightarrow \mathbb{R}^d$, encoding preferred labels, synonyms, definitions, and ontology metadata. An optional Datalog program $\Pi = (\mathcal{F}, \mathcal{R})$ encodes ground facts $\mathcal{F}$ derived from OWL 2 RL axioms and derivable closure rules $\mathcal{R}$ (§3.4).

The alignment relation set is

$$\mathcal{R}_A = \{\text{equivalent}, \text{source_subsumed_by_target}, \text{source_subsumes_target}\}.$$

In description-logic notation: `equivalent` expresses $u \equiv v$; `source\_subsumed\_by\_target` expresses $u \sqsubseteq v$ (the source entity is more specific); `source\_subsumes\_target` expresses $u \sqsupseteq v$ (the source entity is more general). The full reference alignment is

$$A_{\text{ref}} \subseteq \{(u, r, v) \mid u \in V_i, v \in V_j, i \neq j, r \in \mathcal{R}_A\},$$

split into train, validation, and hidden test subsets (§3.8).

2.2 Main Track: Typed Candidate Ranking

The main track defines three ontology-pair tasks:

$$\mathcal{T} = \{\text{NCIT-DOID}, \text{SNOMED-FMA}, \text{SNOMED-NCIT}\}.$$

Each task $T_{ij} : O_i \rightarrow O_j$ aligns entities from source ontology O_i to target ontology O_j .

Query structure and the pool model. The competition adopts a *pool model* in which each source entity $s \in V_i$ may contribute up to two queries: an equivalence query (Q0) and a subsumption query (Q1). This design ensures that equivalence and subsumption prediction are evaluated independently for sources that have both types of reference alignment.

Each query is a triple

$$q = (s_q, O_j, C_q, \text{QueryID}_q),$$

where $s_q \in V_i$ is the source entity, O_j is the target ontology, $C_q \subseteq V_j$ with $|C_q| = 50$ is the fixed candidate set (§3.7), and $\text{QueryID}_q \in \{\text{Q0}, \text{Q1}\}$ identifies the query type. Participants score all *candidate-relation pairs*

$$(v, r) \in C_q \times \mathcal{R}_A,$$

yielding $|C_q| \times |\mathcal{R}_A| = 50 \times 3 = 150$ scored pairs per query (the *block size*). A participant system defines a scoring function

$$f_\theta(s_q, r, v \mid G, X, \Pi),$$

and submits, for each query q , a ranked list

$$\pi_q = [(v_1, r_1), (v_2, r_2), \dots, (v_{150}, r_{150})],$$

ordered by descending score. A prediction (v, r) is *correct* only when both the target entity and the relation type match a reference alignment entry for q . The submission format is described in §5.1.

Three levels of information. The competition exposes three information levels of increasing richness:

Level 1 (Triples). The participant uses G alone: graph structure and training anchors. This level is compatible with standard KG embedding methods and graph neural networks.

Level 2 (Triples + Properties). In addition to G , the participant uses X : preferred labels, synonyms, textual definitions, semantic categories, and source metadata. This level is suited to text-enhanced KG models and biomedical language models.

Level 3 (Triples + Properties + Datalog). The participant additionally uses Π : OWL 2 RL ground facts and derivable closure rules. This level is intended for symbolic, rule-aware, and neuro-symbolic systems.

2.3 Complex Track: OWL Expression Generation

Let $\mathcal{O}_B = \{\text{BFO}, \text{RO}, \text{ChEBI}, \text{CL}, \text{GO}, \text{PATO}, \text{UBERON}, \text{WBbt}\}$ be the *background ontology set*. The three source ontologies are the Human Phenotype Ontology (HP) [Köhler et al., 2021], the Mammalian Phenotype Ontology (MP) [Smith and Eppig, 2009], and the Worm Phenotype Ontology (WBP) [Harris et al., 2014].

For a source concept $c \in V(O_s)$ with $O_s \in \{\text{HP}, \text{MP}, \text{WBP}\}$, the task is to generate an OWL class expression $\hat{\phi}$ drawn from the OWL 2 $\mathcal{EL}$ fragment:

$$\phi ::= A \mid \phi_1 \sqcap \phi_2 \mid \exists r. \phi, \quad (1)$$

where $A \in \bigcup_{O \in \mathcal{O}_B} N_C(O)$ is a named class from a background ontology and $r \in N_R$ is an object property from the BFO or RO vocabularies. The task requires producing a complete owl:equivalentClass axiom for c , serialised in OWL Manchester Syntax [Horridge and Patel-Schneider, 2008].

Worked example. The HP concept *decreased circulating cortisol level* (HP:0008163) has the reference logical definition:

$$\begin{aligned} \exists \text{has_part.} \Big(& \text{decreased_amount} \sqcap \exists \text{inheres_in.} (\text{cortisol} \sqcap \exists \text{part_of. blood}) \\ & \sqcap \exists \text{has_modifier. abnormal} \Big), \end{aligned}$$

where *decreased_amount* and *abnormal* are PATO qualities, *cortisol* is a ChEBI chemical entity, *blood* is a UBERON anatomical structure, and the object properties (*has_part*, *inheres_in*, *part_of*, *has_modifier*) belong to BFO or RO. A correct prediction requires jointly retrieving all relevant named classes from the background ontologies and assembling them into a logically coherent OWL 2 $\mathcal{EL}$ expression, constituting a structured generation problem that rewards retrieval precision and compositional reasoning.

Background property vocabulary. BFO and RO object properties used in complex track definitions include: *part_of* (BFO:0000050), *has_part* (BFO:0000051), *has_quality* (RO:0000086), *has_modifier* (RO:0002573), *inheres_in* (RO:0000052), *located_in* (RO:0001025), *results_in_development_of* (RO:0002296), and *characteristic_of_part_of* (RO:0002314), among others.

3 Dataset Construction

3.1 Source Acquisition

The main track dataset is constructed from five biomedical ontologies (Table 1) together with two reference resources used to derive the ground-truth alignment but not included in the released graph. Two additional reference resources, MONDO [?] and UMLS [Bodenreider, 2004], are used during alignment construction (§3.5) but are not distributed as competition data.

Every source must have a pinned version and a SHA-256 checksum before a production release. Open sources (DOID, NCIT, ORDO, MONDO) are downloaded automatically into a content-addressed

Table 1: Source ontologies in the competition graph (release v0.2.0-rc3).

Ontology	License	Classes	Subclass ax.	Role
SNOMED CT (2026-03-01)	SNOMED CT Affiliate	486,565	2,492,632	task ontology
NCIT	CC BY 4.0	206,085	246,790	task ontology
FMA	FMA License	104,721	104,744	task ontology
ORDO 4.8	CC BY 4.0	14,537	25,665	task ontology
DOID	CC0 1.0	12,127	17,097	task ontology
MONDO	CC BY 4.0	—	—	reference
UMLS 2026AA	UMLS Metath. License	—	—	reference

cache. Restricted sources (SNOMED CT, UMLS) require institutional access or organiser-mediated retrieval; their checksums are recorded privately and their data are never redistributed in public artifacts. Each source write produces a `source_lock.json` recording the version, access mode, checksum, license, and redistribution decision; a companion `source_audit.json` records raw concept counts and axiom counts against the pinned version.

3.2 Normalisation

Each source ontology O_i is normalised into five internal artefacts prior to projection:

- `entities.tsv`: the set of named classes V_i , each with a stable identifier, IRI, ontology tag, and deprecation flag.
- `hierarchy.tsv`: asserted and ELK-inferred direct `SubClassOf` edges (c, c') with an `is_inferred` flag.
- `ontology_axioms.tsv`: OWL axioms beyond simple subclass hierarchy (e.g. `SubClassOf` with a complex filler, `EquivalentClasses`, existential restrictions), retained for downstream Datalog extraction.
- `source_annotations.tsv`: public-safe entity properties (preferred labels, synonyms, definitions, semantic categories, source version) that may appear in the released `properties.csv`.
- `xref_annotations_private.tsv`: exact-match annotations, database cross-references, mapping annotations, and restricted-source fields used as private alignment evidence. These fields are withheld from all public artefacts.

SNOMED CT is handled via its OWL axiom refset when available, or via stated RF2 relationships otherwise, with the projection mode recorded explicitly. NCIT, DOID, ORDO, and FMA are processed as OWL/OBO/RDF ontologies.

3.3 Graph Projection

The graph projection converts the normalised ontology representation into a flat edge table used as $G_i = (V_i, E_i)$.

Hierarchy backbone. Every asserted and ELK-inferred [Kazakov et al., 2014] direct `SubClassOf` edge contributes a `subclass_of` relation. ELK materialises only *direct* inferred subclass edges (i.e., removing transitive redundancies), providing a compact but semantically complete hierarchy. Each edge carries an `is_inferred` flag distinguishing asserted axioms from reasoner-derived entailments.

OWL2Vec* projection. Beyond the hierarchy backbone, the pipeline applies the OWL2Vec* projector [Chen et al., 2021] from the mOWL library [?], configured with `bidirectional_taxonomy=False`, `only_taxonomy=False`, and `include_literals=True`. This projector normalises complex OWL axioms (including existential restrictions, property chains, and annotation assertions) into a flat set of labelled edges, making the resulting graph amenable to standard KG representation learning algorithms without specialised OWL parsers. The projected edge table records, for each edge: `head_id`, `relation`, `tail_id`, the source axiom type, the `projection_method`, the `is_inferred` flag, and provenance metadata.

Cross-ontology anchors. A small set of public training equivalences, generated by normalised label matching and validated against the training reference alignment, are injected as edge entries with `source=anchor`. Training equivalences promoted to anchors are *removed* from the public training label file; anchor information is available only through the graph structure. Validation and test mappings are never promoted to anchors.

Unified graph. The unified competition graph merges all per-ontology projected edge tables:

$$G = \left(\bigcup_i V_i, \bigcup_i E_i \cup E_{\text{anchor}} \right).$$

The released `graph/triples.csv` file encodes this graph with columns `head_id`, `relation`, `tail_id`, `head_ontology`, `tail_ontology`, `source`, `is_inferred`, `is_anchor`, and provenance metadata.

3.4 Datalog Projection

The Datalog layer is a canonical OWL 2 RL [Motik et al., 2012] extraction over stable term identifiers. It is derived directly from the original OWL source documents and their import closure (*not* from the mOWL graph projection), ensuring logical faithfulness to the source axioms.

Predicates. The canonical predicate vocabulary is:

```
source_triple(g:symbol, s:symbol, p:symbol, o:symbol)
triple(s:symbol, p:symbol, o:symbol)
inconsistency(rule_id:symbol, a,b,c,d:symbol)
```

The predicate `source_triple` stores asserted logical RDF triples. The predicate `triple` is the asserted-plus-entailed closure computed by the OWL 2 RL/RDF rule program. The predicate `inconsistency` records OWL 2 RL-detectable constraint violations.

OWL 2 RL compatibility. OWL 2 RL is Horn-rule based: only constructs that admit Horn-rule encoding are supported. Disjunctive constructs such as $A \sqsubseteq B \sqcup C$ are outside OWL 2 RL and are excluded rather than approximated. For example, `ObjectUnionOf` is represented as RDF-list triples with rules deriving membership in the union from membership in any member class; no disjunctive rule head is emitted. Every excluded axiom is written to `reports/unsupported_axioms.csv` with its axiom type and reason for exclusion; in release `v0.2.0-rc3`, the unsupported axiom count is 0.

Legacy named-class projection. For compatibility, the legacy named-class Datalog projection (predicates `subclass`, `edge`, `equiv`, `domain`, `range`, `inverse`, `subprop`, `transitive`) is emitted as `legacy_projection_facts.dl` and `legacy_projection_rules.dl`. In release `v0.2.0-rc3`, the legacy projection encodes 2,782,648 ground facts distributed as: `subclass` 1,647,316; `edge` 1,134,584; `equiv` 10; `domain` 194; `range` 383; `inverse` 38; `subprop` 97; `transitive` 26.

Term dictionary. The file `graph/datalog_terms.tsv` maps stable Datalog symbols back to IRIs, literals, datatypes, language tags, and blank nodes, enabling participants to interpret Datalog conclusions in terms of the original ontology vocabulary.

3.5 Reference Alignment Construction

The reference alignment A_{ref} is the ground-truth set of cross-ontology typed mappings against which all evaluation is performed. Its construction follows three stages: equivalence evidence collection, subsumption derivation, and split assignment.

Equivalence evidence. Equivalence references are assembled from multiple private evidence streams: (i) trusted exact-match annotations (`skos:exactMatch`, `oboInOwl:hasDbXref`) drawn from the source ontologies; (ii) UMLS MRCONSO cross-references, which provide database cross-reference aggregation across clinical vocabularies; (iii) MONDO mappings, which provide

curated cross-ontology equivalences across disease ontologies; and (iv) organiser-curated mappings where automated sources are incomplete or conflicting. All evidence is accumulated in `xref_annotations_private.tsv` and validated before inclusion; the evidence fields are removed from `properties.csv` and all public artefacts before release.

Subsumption derivation. Subsumption references are derived from trusted equivalences and the ELK-augmented target-ontology hierarchy. For a source entity u equivalent to target entity v , and a target-side ancestor a (i.e., $v \sqsubseteq^+ a$), the pipeline derives:

$$(u, \text{source_subsumed_by_target}, a) \in A_{\text{ref}}.$$

Symmetrically, for a target-side descendant d (i.e., $d \sqsubseteq^+ v$):

$$(u, \text{source_subsumes_target}, d) \in A_{\text{ref}}.$$

Derivation depth and the total number of generated subsumption references are capped to avoid flooding the evaluation set with trivial remote ancestors. All subsumption references record their evidence distance and derivation method for auditability.

Curation and split assignment. Splits are assigned by source entity, not by individual reference row: if a source entity has multiple valid target entities or relation types, all are assigned to the same split. This prevents the same source entity from appearing in both training and test labels with different targets, a leakage vector present in naive row-level splits. The pipeline preserves approximate relation balance across train, validation, and test splits for each task; imbalances are recorded in the private split report.

3.6 Pool Model and Per-Query Design

The competition adopts a *pool model* that emits up to two queries per source entity: a Q0 (equivalence-primary) query and a Q1 (subsumption-primary) query. This design separates the evaluation of equivalence and subsumption matching, which are qualitatively distinct problems.

- **Q0 (equivalence query):** The preferred gold pair is the source entity’s equivalence target, with relation `equivalent`. Every source entity with at least one equivalence reference contributes a Q0 query.
- **Q1 (subsumption query):** The preferred gold pair is one of the source entity’s ancestors or descendants in the target ontology, with relation `source_subsumed_by_target` or `source_subsumes_target`. A source entity contributes a Q1 query only when it has a direct subsumption reference that is not derivable from its equivalence reference alone; specifically, a *primary subsumption reference* generated by a direct non-equivalence method.

Under the canonical build (v0.2.0-rc3), the total test query count is 30,216, comprising 15,550 Q0 queries and 14,666 Q1 queries. The same source entity appears on consecutive rows in the candidate and submission files, and the submission is scored positionally: block k of the submission corresponds to row k of `test.cands.tsv`.

3.7 Candidate Generation

Each query $q = (s_q, O_j, C_q, \text{QueryID}_q)$ receives a fixed candidate set C_q of exactly 50 target entities from O_j . Candidate generation follows a tiered strategy ensuring that every candidate set contains: (i) *all gold targets* for q (ensuring the task is solvable); (ii) *structural hard negatives* drawn from the ELK-augmented target-ontology hierarchy neighbourhood of the gold targets (ancestors, descendants, and siblings up to configurable depth), increasing difficulty; (iii) *lexical hard negatives* obtained from near-match results of label-normalised entities from the target ontology; (iv) *semantic hard negatives* using bi-encoder retrieval via MiniLM-L12-v6 to rank a pool of semantically similar candidates compared to the preferred gold’s target label; and (v) *deterministic random fillers* selected from the remaining target entities using a seeded RNG, providing coverage and preventing trivial exclusion strategies. Known-positive targets for the same source are never used as negatives.

The candidate generation strategy can be configured at build time; release v0.2.0-rc3 uses the `mixed_semantic_lexical_then_random` strategy, which fills hard-negative budget with lexical and semantic near-matches before random fillers. Candidate composition statistics are reported in `reports/candidate_metrics.json`.

Table 2: Per-task dataset statistics for release v0.2.0-rc3 (seed=20260426).

Task	Split	Queries		Reference alignments by relation		
		Total	Cands	$\equiv$	$\sqsubseteq$	$\sqsupseteq$
NCIT-DOID	train	7,339	50	3,681	4,826	3,082
	valid	1,207	50	606	803	589
	test	3,683	50	1,849	2,421	1,622
SNOMED-FMA	train	10,568	50	6,117	6,090	6,973
	valid	1,734	50	1,003	1,000	1,095
	test	5,319	50	3,078	3,064	3,433
SNOMED-NCIT	train	41,957	50	21,014	25,079	13,754
	valid	7,095	50	3,554	4,253	2,351
	test	21,214	50	10,623	12,768	7,048
Total test queries		30,216	50	15,550	18,253	12,103

3.8 Data Splits

All three splits (train, validation, test) are constructed simultaneously by the benchmark pipeline, which assigns each source entity to exactly one split across all its reference rows. The default split ratios target approximately 60%/20%/20% (train/valid/test) by source entity count, subject to relation-balance constraints. The train split additionally contributes graph anchors (see §3.3). Table 2 reports per-task, per-split query counts and reference alignment counts for release v0.2.0-rc3.

3.9 Leakage Prevention

The release pipeline applies the following leakage checks before any public artefact is written:

1. No hidden test pair appears in `graph/triples.csv`, `training anchors`, `alignments/train.tsv`, `alignments/valid.tsv`, `graph/facts.dl`, `graph/rules.dl`, or any candidate answer column.
2. Exact-match annotations and xref fields are removed from `graph/properties.csv` and public reports.
3. Training equivalences promoted to graph anchors are removed from `alignments/train.tsv`.
4. Test candidate files (`tasks/<TASK>/test.cands.tsv`) expose only `SrcEntity` and `TgtCandidates` columns; no gold labels are present.
5. Hidden answer SHA-256 checksums are stored privately and verified after evaluation.

A public leakage audit report (`reports/leakage_audit_public.md`) documents the checks performed for each release. The full private audit, including hidden test answer verification, is retained internally.

4 Dataset Statistics

Table 1 reports ontology-level class and subclass-axiom counts. Table 3 reports unified-graph-level statistics for release v0.2.0-rc3. Table 2 reports per-task alignment counts; Table 4 reports complex track statistics.

The complex track reference alignments are the logical definitions already encoded in the source ontologies as `owl:equivalentClass` axioms expressed in terms of the background ontologies. These serve as the ground truth: each definition is a compositional OWL 2 $\mathcal{EL}$ expression of the form given in Equation (1).

Public Release Dataset. The public package (BioKG-Align-kit) includes: `graph/triples.csv`, `graph/properties.csv`, `graph/facts.dl`, `graph/rules.dl`,

Table 3: Unified competition graph statistics (v0.2.0-rc3).

Artefact	Quantity
Nodes ($ V $)	824,035
Hierarchy triples ($ E $, incl. inferred)	2,886,928
Public cross-ontology anchors	10
Datalog ground facts (legacy projection)	2,782,648
of which subclass	1,647,316
of which edge	1,134,584
of which equiv	10
of which domain/range/inverse/subprop/transitive	644
Unsupported OWL 2 RL axioms	0

Table 4: Complex track statistics (v0.2.0-rc3). Splits are determined by the competition organisers.

Source ontology	LDs	Background ($\mathcal{O}_B$)
HP (Human Phenotype Ontology)	5,554	BFO, RO, ChEBI, CL, GO, PATO, UBERON, WBbt
MP (Mammalian Phenotype Ontology)	9,312	
WBP (Worm Phenotype Ontology)	770	
Total	15,636	

Object properties are drawn from BFO and RO for all three source ontologies.

graph/datalog_terms.tsv, schema descriptors, alignments/train.tsv, alignments/valid.tsv, per-task candidate and label files for train and valid splits, test candidate files (no gold labels), evaluation/scorer, evaluation/sample_submission.tsv, baseline prediction files, and statistical reports. The private organiser package additionally contains hidden test answer files, graded relevance files, preferred-pair files, private baseline scores, source-lock audits, and full reference counts.

5 Evaluation Methodology

5.1 Submission Format and Block Scoring

A submission is a TSV file with four columns: SrcEntity, TgtEntity, Relation, Score. Each row assigns a real-valued confidence score to a candidate-relation pair. The submission covers all three task pairs in canonical order (NCIT-DOID, SNOMED-FMA, SNOMED-NCIT) and is structured as *blocks*: for each query (corresponding to one row of the test candidate file in row order), the submission contains exactly $\text{block_size} = |C_q| \times |\mathcal{R}_A| = 50 \times 3 = 150$ consecutive rows. Under the canonical v0.2.0-rc3 build:

$$N_{\text{queries}} = 30,216, \quad N_{\text{rows}} = 30,216 \times 150 = 4,532,400.$$

The scorer walks test.cands.tsv row by row, assigning submission rows to blocks positionally. Within a block, the 150 rows may appear in any order; the scorer indexes by (TgtEntity, Relation). Duplicate pairs within a block are resolved by max-score; missing pairs are assigned score 0.0 (effective last rank). Ties in the final ranking are broken by ascending TgtEntity then by canonical relation order (equivalent $\prec$ source_subsumed_by_target $\prec$ source_subsumes_target).

Fatal errors (wrong total row count, block-entity mismatch, invalid relation string, non-parseable score) reject the submission immediately. Warnings (duplicate pairs, missing pairs) are diagnostics that allow scoring to proceed.

5.2 Preferred-Pair Family

For each query q , the organiser designates exactly one *preferred gold pair*

$$p_q^* = (v_q^*, r_q^*) \in C_q \times \mathcal{R}_A.$$

If an equivalence reference is available among the query’s gold pairs, it is selected as preferred; otherwise, the primary subsumption direction is used. The *Preferred Typed Reciprocal Rank* for query q is

$$\text{PrefTRR}(q) = \frac{1}{\text{rank}_q(p_q^*)},$$

where $\text{rank}_q(p_q^*)$ is the one-indexed position of p_q^* in the ranked submission block for q . A system receives non-zero credit only when both target entity *and* relation type exactly match the preferred gold pair. The per-task macro-average is

$$\text{PrefTypedMRR}(T_i) = \frac{1}{|Q_i|} \sum_{q \in Q_i} \text{PrefTRR}(q),$$

and the **primary leaderboard metric** is the macro-average across all tasks:

$$\text{MacroPrefTypedMRR} = \frac{1}{|\mathcal{T}|} \sum_{T_i \in \mathcal{T}} \text{PrefTypedMRR}(T_i). \quad (2)$$

The preferred-pair family additionally reports PrefTypedH@K for $K \in \{1, 5, 10\}$:

$$\text{PrefTypedH@K}(q) = \mathbf{1}[\text{rank}_q(p_q^*) \leq K].$$

Relation Macro-F1 on the preferred entity. A diagnostic sub-metric evaluates relation-type classification conditioned on entity retrieval. The system’s top-ranked entity $\hat{v}_q$ is determined by taking the maximum score across all three relations for each candidate. Queries for which $\hat{v}_q \neq v_q^*$ are excluded from this metric (marked *unmeasurable*). For entity-correct queries, the predicted relation is the one paired with $\hat{v}_q$ in the top-1 submission row; macro-F1 is computed across the three relation types. This metric isolates the difficulty of relation typing from the difficulty of entity retrieval. The support (number of entity-correct queries) must always be reported alongside the F1 value.

5.3 Hierarchy-Aware Typed nDCG@10

The *Hierarchy-Aware Typed nDCG@10* (H-nDCG@10) metric extends standard nDCG [?] with a graded relevance gain table that awards partial credit to hierarchically close ($\text{TgtEntity, Relation}$) pairs. This metric is the **secondary leaderboard metric**.

Gain table construction. For each query q with preferred gold pair (v^*, r^*) , the gain function $g_q : (C_q \times \mathcal{R}_A) \rightarrow [0, 1]$ is defined as follows. At distance $d = 0$ (the gold target itself):

Preferred (v^*, r^*)	$g_q(v^*, \equiv)$	$g_q(v^*, \sqsubseteq)$	$g_q(v^*, \sqsupseteq)$
$(v^*, \equiv)$	1.0	0.6	0.6
$(v^*, \sqsubseteq)$	0.0	1.0	0.0
$(v^*, \sqsupseteq)$	0.0	0.0	1.0

The 0.6 entries in the equivalence-preferred row capture near-miss credit: a system that correctly identifies the gold entity but predicts the wrong subsumption direction receives partial credit. At hierarchical distance $d \in \{1, 2, 3\}$ (ancestors and descendants of v^* in the ELK-augmented target-ontology hierarchy, restricted to entities in C_q):

Preferred r^*	Walk	Pair	Gain
$\equiv$	ancestors	$(\text{anc}, \sqsubseteq)$	$\frac{0.6}{d+1}$
$\equiv$	descendants	$(\text{desc}, \sqsupseteq)$	$\frac{0.6}{d+1}$
$\sqsubseteq$	ancestors	$(\text{anc}, \sqsubseteq)$	$\frac{1.0}{d+1}$
$\sqsupseteq$	descendants	$(\text{desc}, \sqsupseteq)$	$\frac{1.0}{d+1}$

The walk depth is capped at $\max_d = 3$, giving a minimum gain of $\frac{0.6}{4} = 0.15$ for an equivalence-preferred query at maximum depth. Gains for pairs not in the above table are zero.

Metric computation. Let p_i^q be the i -th pair in the ranked submission for query q . The Discounted Cumulative Gain at K is

$$\text{DCG}@K(q) = \sum_{i=1}^K \frac{g_q(p_i^q)}{\log_2(i+1)}.$$

The Ideal DCG is

$$\text{IDCG}@K(q) = \sum_{i=1}^K \frac{g_q^{(i)}}{\log_2(i+1)},$$

where $g_q^{(1)} \geq g_q^{(2)} \geq \dots$ are the gains in $\{g_q(v, r) : (v, r) \in C_q \times \mathcal{R}_A\}$ sorted in descending order. The normalised DCG is

$$\text{nDCG}@K(q) = \frac{\text{DCG}@K(q)}{\text{IDCG}@K(q)}, \quad \text{with } \text{nDCG}@K(q) = 0 \text{ when } \text{IDCG}@K(q) = 0.$$

H-nDCG@10 is the macro-average of nDCG@10(q) across all queries.

5.4 Complex Track Evaluation: GED-Based F-Measure

The complex track is evaluated using a graph edit distance (GED) approach following Silva et al. [2025]. Each OWL 2 $\mathcal{EL}$ class expression ϕ is converted to a directed labelled graph $\mathcal{G}(\phi)$ whose nodes represent named classes and existential restrictions and whose edges encode the OWL constructor hierarchy. The GED between a generated expression $\hat{\phi}$ and the reference expression ϕ^* is computed using the NetworkX library [?] under unit node and edge substitution, insertion, and deletion costs.

Each generated mapping is assigned to one of six outcome categories:

Category	Criterion
<i>correct</i>	$\mathcal{G}(\hat{\phi}) = \mathcal{G}(\phi^*)$
<i>contains reference</i>	$\mathcal{G}(\phi^*)$ is a subgraph of $\mathcal{G}(\hat{\phi})$
<i>contained in reference</i>	$\mathcal{G}(\hat{\phi})$ is a subgraph of $\mathcal{G}(\phi^*)$
<i>incorrect</i>	none of the above
<i>false positive</i>	no reference exists for c ; mapping was generated
<i>false negative</i>	reference exists for c ; no mapping was generated

These categories map to precision, recall, and F-measure following the category-to-outcome rules defined in Silva et al. [2025]. The primary complex track metric is the macro-averaged F_1 across source ontologies:

$$F_1^{\text{macro}} = \frac{1}{3}(F_1^{\text{HP}} + F_1^{\text{MP}} + F_1^{\text{WBP}}).$$

Submitted expressions must be parseable by `xml.etree.ElementTree`. The complex track leaderboard is independent of the main track leaderboard.

6 Baseline Systems and Results

All baselines are implemented in the `biokg_align` Python package using PyKEEN [Ali et al., 2021] for KGE models. Each baseline is trained or applied on the same public graph, node properties, and training alignment, and evaluated on the hidden test split.

Random. A lower-bound baseline that samples scores uniformly at random from $[0, 1)$ for each candidate-relation pair. Provides a calibration anchor for all other methods.

Exact lexical. Scores candidate-relation pairs using exact string matching after label normalisation (lowercasing, punctuation removal, whitespace normalisation). Returns score 1.0 for an exact normalised-label match, else 0.0. All three relation types receive identical entity scores; relation type is determined by a fixed prior.

Hybrid lexical. Scores entities using a weighted combination of exact and approximate string similarity over preferred labels, synonyms, and definitions (token overlap, character-level similarity, BM25-style TF-IDF weighting). Relation types are assigned by a corpus-level prior over training references. This is the strongest lexical-only baseline and serves as a text-feature ceiling for methods that do not exploit graph structure.

Word embedding-based. Encodes each entity as a sentence embedding of its concatenated preferred label, synonyms, and definition using a pre-trained biomedical language model (e.g., BioBERT or a sentence-BERT variant). Scores candidate-relation pairs by cosine similarity between source and candidate embeddings. Relation types are assigned by the same corpus-level prior as the hybrid lexical baseline.

KGE baselines (PyKEEN). Four knowledge graph embedding models are trained on the released graph using PyKEEN, with shared hyperparameters: embedding dimension 64, early stopping based on validation MRR (patience 3, frequency 10 epochs, relative delta 0.002). The models are:

TransE [Bordes et al., 2013] models relations as translations in embedding space: $\mathbf{e}_h + \mathbf{r} \approx \mathbf{e}_t$ for a true triple (h, r, t) . The scoring function is $f(h, r, t) = -\|\mathbf{e}_h + \mathbf{r} - \mathbf{e}_t\|_2$.

DistMult [Yang et al., 2015] uses a bilinear scoring function $f(h, r, t) = \mathbf{e}_h^\top \text{diag}(\mathbf{r}) \mathbf{e}_t$, constraining relation matrices to be diagonal. DistMult is symmetric in head and tail, which is a structural mismatch for directed subsumption relations; nevertheless, the model benefits from the informativeness of the multi-relational training signal.

Complex [Trouillon et al., 2016] extends DistMult to the complex number field. Entity and relation embeddings are complex-valued: $\mathbf{e}_h, \mathbf{r}, \mathbf{e}_t \in \mathbb{C}^d$. The scoring function is

$$f(h, r, t) = \text{Re}(\langle \mathbf{e}_h, \mathbf{r}, \bar{\mathbf{e}}_t \rangle) = \text{Re} \left(\sum_{k=1}^d e_{h,k} \cdot r_k \cdot \overline{e_{t,k}} \right),$$

where $\bar{\mathbf{e}}_t$ denotes the complex conjugate of $\mathbf{e}_t$. The use of complex conjugate breaks the symmetry of DistMult, enabling Complex to model asymmetric relations such as directional subsumption ($\sqsubseteq$, $\sqsupseteq$). This makes Complex theoretically well suited to the BioKG-Align task, where correct relation typing requires distinguishing `source_subsumed_by_target` from `source_subsumes_target`. In practice, Complex can be seen as a factorisation of the full triple relation matrix over complex space, with universal approximation properties for the class of relations that can be expressed as linear combinations of complex-valued rank-1 matrices. Training uses the Complex loss (negative sampling with binary cross-entropy), with complex-number arithmetic handled natively by PyKEEN’s implementation.

RotatE [Sun et al., 2019] represents relations as element-wise rotations in the complex unit circle: $e_{h,k} \cdot r_k = e_{t,k}$, where $|r_k| = 1$. This constraint models composition, symmetry, antisymmetry, inversion, and transitivity as geometric operations on the complex plane, and has strong theoretical properties for capturing hierarchical and directional relations. The scoring function is $f(h, r, t) = -\|\mathbf{e}_h \circ \mathbf{r} - \mathbf{e}_t\|_2$.

Baseline results. Table 5 reports macro-averaged results across the three main track tasks on the test split for release v0.2.0-rc3 (30,216 test queries, 50 candidates per query). Figure-level analysis reveals three key patterns: (i) lexical and embedding methods achieve the highest Preferred Typed MRR (0.423–0.427), reflecting their effectiveness for equivalence matching where label similarity is a strong signal; (ii) among KGE models, RotatE achieves the best overall H-nDCG@10 (0.203), benefiting from its ability to model directional and hierarchical relations; (iii) lexical methods have near-zero macro-F1 for non-equivalence relations, indicating they fail entirely at subsumption typing, a critical gap that more sophisticated methods must address; and (iv) RotatE achieves the highest Relation Macro-F1 on the preferred entity (0.547), confirming that embedding geometry that respects directionality transfers to relation typing.

Table 6 reports per-task breakdown for selected baselines to illustrate task difficulty variation.

Key observations. The gap between the best entity-only MRR (0.519 for word embedding) and the best Preferred Typed MRR (0.427) quantifies the additional difficulty of relation typing: methods

Table 5: Macro-averaged baseline results on the test split (v0.2.0-rc3, 30,216 queries). **PrefMRR**: Macro Preferred Typed MRR (primary metric). **H-nDCG**: Hierarchy-Aware Typed nDCG@10 (secondary metric). **RelF1**: Relation Macro-F1 on preferred entity. **EntMRR**: entity-only MRR (ignoring relation type).

Baseline	PrefMRR	H-nDCG	PrefH@1	PrefH@5	PrefH@10	EntMRR	EntH@1	RelF1
Random	0.037	0.045	0.007	0.035	0.068	0.091	0.021	0.313
TransE	0.088	0.078	0.027	0.109	0.190	0.166	0.062	0.335
DistMult	0.117	0.157	0.037	0.160	0.274	0.209	0.075	0.455
ComplEx	0.143	0.152	0.055	0.208	0.320	0.244	0.127	0.449
RotatE	0.187	0.203	0.108	0.250	0.330	0.236	0.125	0.547
Exact lexical	0.061	0.052	0.037	0.056	0.085	0.102	0.043	0.230
Hybrid lexical	0.423	0.381	0.353	0.503	0.561	0.516	0.431	0.230
Word embedding	0.427	0.376	0.354	0.500	0.569	0.519	0.419	0.230

Table 6: Per-task baseline results (v0.2.0-rc3), Preferred Typed MRR | H-nDCG@10.

Baseline	NCIT-DOID		SNOMED-FMA		SNOMED-NCIT	
Random	0.037	0.045	0.038	0.045	0.038	0.046
DistMult	0.098	0.143	0.124	0.161	0.128	0.169
ComplEx	0.140	0.152	0.133	0.129	0.155	0.176
RotatE	0.200	0.207	0.170	0.171	0.191	0.230
Hybrid lexical	0.419	0.373	0.432	0.398	0.418	0.373
Word embedding	0.430	0.386	0.408	0.353	0.442	0.389

that correctly retrieve the target entity still fail to predict the correct relation type in approximately 18% of cases. Lexical and embedding baselines collapse on relation typing (Relation Macro-F1 = 0.230), which is the prior-only score; their relation predictions are driven entirely by training priors and not by any per-query signal. By contrast, RotatE achieves Relation Macro-F1 = 0.547, indicating that geometric embedding of the multi-relational graph transfers meaningfully to relation-type discrimination. The current best Preferred Typed MRR of 0.427 leaves substantial room for methods that combine textual signals with graph structure, logical rules, or both.

7 Reproducibility and Competition Infrastructure

The benchmark pipeline is implemented in the `biokg-align` Python package (managed with Poetry). All build steps (source acquisition, normalisation, graph projection, Datalog extraction, alignment construction, candidate generation, baseline training, and evaluation) are reproducible from a single configuration file (`configs/fixture.json` or release-mode equivalent) and a random seed (release v0.2.0-rc3 uses seed 20260426). Baseline training uses PyKEEN with fully seeded RNG states. The scorer is a deterministic CLI tool distributed with the starting kit.

The competition platform is CodaBench; data are distributed via Hugging Face Datasets. The starting kit includes parsers for all data formats, the official scorer, submission validation utilities, and example submission files. GPU-based baselines are reproducible via SLURM scripts in the `slurm/` directory. CPU-only reproducibility is also supported; KGE baselines automatically fall back to CPU when CUDA is unavailable.

Per the benchmark reproducibility protocol, each baseline run records: the exact command line, configuration file, git commit, source data release version, random seed, hardware summary, prediction files, and metric summaries (JSON and Markdown). This information is included in the released `baseline_predictions/` and `baseline_results*` artefacts.

8 Conclusion

This whitepaper provides the full technical specification for the BioKG-Align competition, grounding each design choice in formal definitions and reporting empirical results on the v0.2.0-rc3 dataset release. The benchmark exposes ontology matching as typed link prediction over a large,

multi-ontology biomedical knowledge graph, with progressively richer information layers enabling participation from diverse methodological backgrounds. The baseline analysis demonstrates that the typed alignment problem is substantially harder than untyped entity retrieval, that lexical and embedding methods fail at relation typing, and that geometric KGE methods show promise for subsumption direction prediction, collectively defining an open research agenda for graph learning, language model-based, rule-aware, and neuro-symbolic methods on ontology-derived biomedical knowledge graphs.

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